

Problem F. Magic

Time Limit 1000 ms

Mem Limit 262144 kB

OS Windows

Did you ever dream of becoming a wizard? Or of being able to solve every problem of every contest?

Valya wanted to try out magic since childhood, so he became a programmer. One time he fell asleep trying to solve a problem, and he saw some magical dreams.

In his dream, Valya owns A red lands, B blue lands and C blue-red lands. A red land gives him one red mana, a blue land — one blue mana, and a blue-red land gives him either one red or one blue mana (for each blue-red land Valya can decide whether he gets red or blue mana from this land). Valya wants to cast a spell that costs X red mana, Y blue mana, and additionally Z mana of any color. Casting this spell will help him to solve the problem he fell asleep to. Determine if Valya has enough mana to use the spell.

Input

The first line contains three space-separated integers A , B and C — the number of red, blue and blue-red lands respectively ($0 \leq A, B, C \leq 10^9$).

The second line contains three space-separated integers X , Y and Z — amount of red mana, blue mana and additional mana of any color required to cast the spell ($0 \leq X, Y, Z \leq 10^9$).

Output

If Valya is able to cast the spell, print “It is a kind of magic” (without quotes).

Otherwise, print “There are no miracles in life” (without quotes).

Sample

Input	Output
3 3 1 2 2 1	It is a kind of magic

Input	Output
5 6 5 6 5 6	There are no miracles in life